Object Oriented Programming

Comprehensive Placement Interview Notes

Ace the Technical Rounds

Chapter 1: Introduction to OOP and Java Fundamentals

What is Object-Oriented Programming?

OOP is a programming paradigm based on the concept of **"objects"**, which contain data (attributes) and code (methods). Unlike procedural programming which focuses on functions, OOP focuses on the real-world entities that the software is trying to represent.

Objects and Classes

In Java, you cannot have an object without a class. Think of it this way:

- **Class:** A *Blueprint* on paper. It defines what a house should look like (rooms, windows, height).
- **Object:** The *Actual House* built on a plot of land. It has real walls you can touch and a real address in memory.

What is a Class?

A Class is a **user-defined data type**. It acts as a logical template/blueprint for creating objects. It consists of:

- **Fields (Attributes):** What the object **knows** (e.g., `color`, `speed`).
- **Methods (Behaviors):** What the object **does** (e.g., `accelerate()`, `brake()`).

What is an Object?

An Object is a **physical instance** of a class. When created, memory is allocated in the **Heap**.

- **State:** Represented by the values of its attributes.
- **Behavior:** Represented by its methods.
- **Identity:** A unique ID/Address used by the JVM.

Code Example: Practical Implementation

Imagine a class `Smartphone`. From this one class, we can create an iPhone object or a Samsung object.

```java
// The Class (Blueprint)
class Smartphone {
    String brand;
    int ram;

    void ring() {
        System.out.println(brand + " is ringing...");
    }
}

// Creating Objects (Instances)
public class Main {
    public static void main(String[] args) {
        Smartphone p1 = new Smartphone(); // Object 1
        p1.brand = "Apple";

        Smartphone p2 = new Smartphone(); // Object 2
        p2.brand = "Samsung";

        p1.ring(); // Output: Apple is ringing...
    }
}
```

1.1.1 Detailed Comparison: Object vs Class

OBJECT	CLASS
Object is an instance of a class.	Class is a blueprint from which objects are created.
Object is a real world entity such as chair, pen, table, laptop etc.	Class is a group of similar objects.
Object is a physical entity.	Class is a logical entity.
Object is created many times as per requirement.	Class is declared once.
Object allocates memory when it is created.	Class doesn't allocate memory when it is created.
Object is created through `new` keyword. *Example:* `Employee ob = new Employee();`	Class is declared using `class` keyword. *Example:* `class Employee{}`
There are different ways to create objects in Java: `new` keyword, `newInstance()` method, `clone()` method, and deserialization.	There is only one way to define a class, i.e., by using the `class` keyword.

Table 1: Key differences between Objects and Classes for Java Interviews.